

Program9: Organize Kubernetes pods using labels and namespace

Project Structure:

Pgm9

->namespaces.yaml

->pods.yaml

Step1: Create pods.yaml

apiVersion: v1

kind: Pod

metadata:

name: frontend-pod

namespace: dev

labels:

app: frontend

env: dev

spec:

containers:

- name: frontend
image: nginx
ports:
- containerPort: 80

Step2: create namespace.yaml

apiVersion: v1

kind: Pod

metadata:

name: backend-pod

namespace: dev

labels:

app: backend

env: dev

spec:

containers:

name: backend
image: nginx
ports:

containerPort: 80

apiVersion: v1
kind: Pod
metadata:
name: frontend-pod
namespace: prod
labels:
app: frontend
env: prod

spec:

containers:

- name: frontend
image: nginx
ports:
- containerPort: 80

Step 3: Apply namespace and pods

apiVersion: v1
kind: Namespace
metadata:
name: dev

apiVersion: v1
kind: Namespace
metadata:
name: prod

```
1rv24mc058_kumaraswamys@jarvis:~$ kubectl apply -f pods.yaml  
  
namespace/dev created  
namespace/prod created
```

Step 4: Verify pods in each namespace

```
1rv24mc058_kumaraswamys@jarvis:~$ kubectl apply -f namespace.yaml  
pod/frontend-pod created  
pod/backend-pod created  
pod/frontend-pod created  
pod/backend-pod created
```

Step 5: Filter pods by label

```
frontend-pod 1/1 Running 0 41s  
1rv24mc058_kumaraswamys@jarvis:~$ kubectl get pods -n prod -l app=frontend  
NAME READY STATUS RESTARTS AGE  
frontend-pod 1/1 Running 0 2m30s
```

```
pod/backend-pod created  
1rv24mc058_kumaraswamys@jarvis:~$ kubectl get pods -n dev  
NAME READY STATUS RESTARTS AGE  
backend-pod 1/1 Running 0 41s  
frontend-pod 1/1 Running 0 41s
```